

EMATM0049 (TIBS) Theme 0: Don't Believe The Hype!

(Teaching block 1 2025/26)

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This is a unit in which there are no right or wrong opinions; the material covered in this unit definitely doesn't tell you what to think, but it is intended to make you think, and it is intended to develop your powers of how to think, to sharpen the tools in your mental toolbox.

We want you to come to all of the reading on this unit with an air of constructively critical scepticism: don't unthinkingly believe everything you read; treat all arguments as testable, as open to question. And, to encourage that, we start this unit with a "preamble" theme, which is about questioning the claims made by other authors. In Theme 0: Don't Believe the Hype, we will draw primarily from the books by Fenn & Raskino (2008); Ritchie (2020); Rosenzweig (2007); and West & Bergstrom (2020).

Rosenzweig's text is something of a classic whereas the Fenn & Raskino text is included here more for completeness than because it has any enduring value: the idea of the "Hype Cycle" is worth knowing about because it is commonly mentioned, but (as we'll see in the seminar session) it is an idea that itself seems to be somewhat over-hyped. The recent books by Ritchie and by West & Bergstrom are both written for a popular audience: each of them is not heavily technical, and can be read in a day or so; they're both strongly recommended for their coverage of bad practice in data science, and in science more generally. Ritchie's book in particular covers the replication crisis that has struck multiple fields in recent years, and which remains an ongoing issue of concern.

After this theme, with our bullshit detectors fully activated, we then proceed to the nine themes that make up the core content of the unit. The title of this unit is Technology, Innovation, Business, and Society because that makes a nice acronym, TIBS, so after this zeroth theme we'll start with Theme 1: Technology Innovation.

27 items

Mastering the hype cycle : how to choose the right innovation at the right time, by Jackie Fenn; Mark Raskino, 2008

[Book](#) | [Further](#)

Science fictions : exposing fraud, bias, negligence and hype in science, by Stuart Ritchie, 2020

[Book](#) | [Further](#)

The halo effect-- and the eight other business delusions that deceive managers, by Philip M. Rosenzweig, 2007

[Book](#) | [Further](#)

Calling bullshit : the art of scepticism in a data-driven world, by Carl T. Bergstrom; Jevin D. West, 2020

[Book](#) | [Further](#)

Reading all four of those books would probably take you much longer than the 12-13 hours we expect you to spend on working through each week's learning materials. So instead we have assembled a priority-ordered list of reading materials for you to work through in sequence, top to bottom...

1. A written transcript of an interview with Phil Rosenzweig, talking about his book *The Halo Effect* (the transcript includes a v brief summary of the book):

The Halo Effect and the Challenge of Management Inquiry: A Dialog Between Phil Rosenzweig and Paul Olk, by Paul Olk; Phil Rosenzweig

[Article](#) | [Essential](#)

2: This video on Gartner's "Hype Cycle" is mercifully brief: it will only waste 4m08s of your life. Try to think of it as time invested.

Mastering the Hype Cycle Book, by Gartner, 2008-10-01T05:40:17-07:00

[Audio-visual document](#) | [Essential](#)

3: This critical review by Dedehayir & Steinert (2006) of *The Hype Cycle* was published with the benefit of several years of hindsight...

The hype cycle model: A review and future directions - in *Technological Forecasting and Social Change*, by Ozgur Dedehayir; Martin Steinert, 2016-07

[Article](#) | [Essential](#)

4: Rosalind Arden's very nice review of Ritchie's "Science Fictions" in the Quillette website is here:

'Science Fictions' Review: Begone, Science Swindlers, by Rosalind Arden, 21 August 2020

[Webpage](#) | [Essential](#)

5: A 43m conversation between Tom Whipple (Science Editor at The Times) and and Stuart Ritchie, author of *Science Fictions*, on the Intelligence-Squared podcast:

Should We Trust Scientists?, by Intelligence Squared, 2020-07-14T10:47:38-07:00

[Audio-visual document](#) | [Essential](#)

6. Bergstrom & West, the authors of *Calling Bullshit*, have produced a series of 10 lectures,

each broken into a sequence of short videos, to accompany their book. The videos are freely available on Youtube: please watch **all** of them!

Calling Bullshit — Video lectures, by Carl Bergstrom; Jevin West, 2017

[Webpage](#) | **Essential**

Background Reading

Here are some further materials that you might want to work through to expand your knowledge of this theme.

Jonathan Pruitt was a young behavioural ecologist with a "superstar" track-record of peer-reviewed publications and prestigious research-grant awards. In January 2020 one of Pruitt's co-authors, Kate Laskowski from the University of California at Davis, published a tweet on Twitter that sparked a "twitterstorm": Laskowski alleged that she had found serious irregularities in experiment data that Pruitt had supplied to her for one of their co-authored papers, and the irregularities were impossible to explain (she had asked Pruitt, he could not explain them) so she asked for their co-authored paper to be withdrawn by the publisher. This prompted other of Pruitt's co-authors to check the data he had sent them, and they found similar irregularities, and they also asked for their co-authored papers to be withdrawn from the record. Jonathan Pruitt hired lawyers who wrote to various of his (now ex-)co-authors, warning them to be very careful in what they said about Pruitt (the implication presumably being that Pruitt might launch a lawsuit against them) this is a highly unusual move among co-authors on scientific papers). And Pruitt's employer, McMaster University in Canada, initiated a slow and confidential inquiry; at the same time, the University of Tennessee at Knoxville, where Pruitt had studied for his PhD, withdrew his PhD thesis from their library collection. The Nov.2021 article in Science by Elizabeth Pennisi gave an update on the story; by August 2022, Pruitt had resigned his professorship and was reportedly working as a high-school science teacher in Florida, as reported in the Nature article by Max Kozlov. Finally in May 2023 the McMaster University enquiry published its findings, as reported in Science by Claudia López Lloreda.

As data probe concludes, spider biologist placed on leave, has Ph.D. thesis 'withdrawn', by Elizabeth Pennisi, 15 November 2021

[Article](#) | **Recommended**

How a scandal in spider biology upended researchers' lives - in Nature, by Max Kozlov, 2022-08-25

[Article](#) | **Recommended**

University investigation found prominent spider biologist fabricated, falsified data | Science | AAAS

[Webpage](#) | **Recommended**

A Financial Times article by Robin Wrigglesworth in Nov.2021 describes how Campbell Harvey, a prominent Professor of Finance, was arguing that there is a replication crisis in Finance too...

The hidden 'replication crisis' of finance - in The Financial Times, by Robin Wigglesworth, 2021

[Article](#) | Recommended

If you are wondering whether the replication crisis affects Data Science or not, two articles worth reading are by William Douglas Heaven writing in MIT Technology Review in Nov.2020; and Luciano Abriata in Towards Data Science in January 2024.

AI is wrestling with a replication crisis | MIT Technology Review, by William Douglas Heaven, November 12, 2020

[Webpage](#) | Recommended

How Artificial Intelligence Might be Worsening the Reproducibility Crisis in Science and Technology | Towards Data Science, by Luciano Abriata, 3 January 2024

[Webpage](#) | Recommended

EMATM0049 (TIBS) Theme 1: Technology & Innovation (Teaching block 1 2025/26)

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We kick off with considering technology and innovation, first taking a broad historical sweep via the writing of David Edgerton; then exploring Carlota Perez's argument that there are cyclical patterns that repeat in successive waves of innovative technology stretching back to the original industrial revolution of the 1760s; and finally we review a hugely influential idea, Clayton Christensen's technical notion of "disruptive technologies" -- and how they can make life difficult for innovators. The books we cover in "Theme 1: Technology and Innovation", include Christensen (1997/2016), Edgerton (2011), and Perez (2002). While we don't require you to read every single book mentioned in the lists of TIBS readings, we really do recommend reading all of "The Innovator's Dilemma" -- it's widely regarded as one of the greatest business books of all time. It should take you about six hours or so.

31 items

Shock Of The Old : Technology and Global History since 1900, by David Edgerton, 2011

[Book](#) | Recommended

The innovator's dilemma : when new technologies cause great firms to fail, by Clayton M. Christensen, 2016

[Book](#) | Recommended

Technological revolutions and financial capital : the dynamics of bubbles and golden ages, by Carlota Perez, 2002

[Book](#) | Recommended

Here is a video lecture (in two parts) made by Dave Cliff to explain this week's theme. A PDF of the powerpoint slides for this video lecture is available in the "Session Notes" section of the Blackboard page for this theme, along with a PDF of the powerpoint slides written by James Pope to introduce the Theme 1 seminar sessions in 2025.

TIBS Theme 1 introduction/overview Part A, by Dave Cliff

[Webpage](#) | Essential | (Video lecture by Dave Cliff)

TIBS Theme 1 introduction/overview Part B, by Dave Cliff

[Webpage](#) | Essential | (Video lecture by Dave Cliff)

Once you have watched the introduction/overview video, please then work your way through the following priority-ordered list of learning resources...

1. A nicely-made short video summary of Christensen's The Innovator's Dilemma by Book Video Club (4m17s):

"The Innovator's Dilemma" by Clayton Christensen - VIDEO BOOK SUMMARY, by Snackable Idea, 2015-03-13T07:31:17-07:00

Audio-visual document | **Essential**

2: The original HBR article on disruptive technologies is: J. Bower & C. Christensen (1995) Disruptive Technologies: Catching the Wave. Harvard Business Review, Jan-Feb 1995, pp.43-53.

Harvard Business Review, by Clayton Christensen; Joseph Bower, 1995

Journal | **Essential**

3: Clayton Christensen gave a one-hour lecture about his work at the Said Business School, University of Oxford (59m14s):

Clayton Christensen on disruptive innovation Clarendon Lectures 10th June 2013, by alan yong, 2017-05-09T05:32:35-07:00

Audio-visual document | **Essential**

4: In June 2014 Jill Lepore, a fellow professor and colleague of Christensen's at Harvard Business School, wrote a critical article for The New Yorker about Christensen's work: The Disruption Machine; What the gospel of innovation gets wrong:

The Disruption Machine: What the Gospel of Innovation Gets Wrong | The New Yorker, by Jill Lepore

Webpage | **Essential**

5: This January 2022 article by Julian Birkinshaw in the Harvard Business Review presents evidence from his research which casts doubt on the prevalence of disruptive innovation in the internet era. It includes an audio version you can listen to (24m56s).

How Incumbents Survive and Thrive, by Julian Birkinshaw

Webpage | **Essential**

6: Haydn Shaughnessy's article in Forbes, discussing the dispute between Christensen and Lepore, among other things:

What Did The Innovator's Dilemma Get Wrong?, by Haydn Shaughnessey

Webpage | **Essential**

7. Video interview with David Edgerton, talking about The Shock of the Old and related matters, in 2017 (23m27s):

Digital Transformation: Interview with David Edgerton, King's College London, by Manuel Stagars, 2017-06-19T20:51:20-07:00

Audio-visual document | **Essential**

8. Carlota Perez, speaking in 2017, explains how the start of the ICT revolution arose in the midst of the Mass-Production revolution, and how the next revolution will be starting mid-way through the ICT revolution: here she says the end of the ICT revolution is 15-30 years away (8m29s).

Carlota Perez: In the midst of ICT revolution; next revolution 30 years out sd STREAM

(2011), by Carlota Perez, 2017-03-12T05:54:43-07:00

[Audio-visual document](#) | Essential

9. A brief review by David Goldblatt in The Independent of David Edgerton's The Shock of the Old:

The Shock of the Old, by David Edgerton, by David Goldblatt, 2007

[Webpage](#) | Essential

10. Video of Carlota Perez giving a monologue at the Ellen MacArthur Foundation in 2018; her opening words are a nice summary of the major surges (18m47s):

The Fifth Industrial Revolution is Happening - Is it Time to Reshape Our Future? | Summit 2018, by Carlota Perez, 2018-07-09T09:34:24-07:00

[Audio-visual document](#) | Essential

11. A longer review of Edgerton's The Shock of the Old, by Steven Shapin in The New Yorker:

What Else Is New? | The New Yorker, by Steven Shapin

[Webpage](#) | Essential

12. Video interview of Carlota Perez speaking in 2021: the first 4m30s are a nice introduction to her analysis; the remainder of the discussion talks about sustainability (with occasional reference to the fashion industry -- this video is produced by Business of Fashion).

The Green Global Age of the Information Revolution: Carlota Perez | #BoFVOICES 2020, by Carlota Perez, 2021-04-02T03:38:51-07:00

[Audio-visual document](#) | Essential

EMATM0049 (TIBS) Theme 2: Beyond Innovation (Teaching block 1 2025/26)

[View Online](#)

After you've come up with an innovation, what next? In this theme, we explore thinking on the processes that unfold after an innovative idea has been come up with. If the innovation has been developed in a commercial context, then there is very often an expectation that it should be commercialized in some sense, to generate ongoing profit beyond the R&D and commercialization costs. This process of "knowledge transfer" or "execution" can benefit from an understanding of how innovative ideas, technologies, or products spread through a population of potential users, of potential customers. In this theme we explore the widely-known work of Geoffrey Moore on "crossing the chasm"; we look at the scholarly roots of Moore's "chasm", which lie in the classic seminal work of Everett Rogers on the diffusion of innovations, and we cover the work of Dartmouth Business School professor Vijay Gorindarajan on execution of innovation.

18 items

The other side of innovation : solving the execution challenge, by Vijay Govindarajan; Chris Trimble, 2010

[Book](#) | Essential

Crossing the chasm : marketing and selling disruptive products to mainstream customers, by Geoffrey A. Moore, 2014

[Book](#) | Further

Diffusion of innovations, by Everett M. Rogers, 2003

[Book](#) | Further

Here is a set of slides by James Pope that were used as an introduction/overview to the seminar session in 2024:

James Pope's slide-deck for TIBS Theme 2 (2024), by James Pope, September 2024

[Webpage](#) | Recommended

As preparation for the seminar discussions, please work your way through this priority-ordered list of learning resources, in sequence:

1. Vijay Govindarajan talking-head video summarising/promoting hi book The Other Side of Innovation (7m07s):

The Other Side of Innovation: Solving the Execution Challenge, by Vijay Govindarajan, 30 March, 2011

[Webpage](#) | Essential

2. Geoffrey Moore giving a retrospective lecture 25 years after he first published his

landmark book *Crossing the Chasm*. The quality of the audio and video could be better, but hearing Moore's perspective on his own widely-known work is invaluable (1h04m36s).

Crossing the Chasm by Geoffrey Moore - Lean Product Meetup, by Geoffrey Moore, 6 March 2015

[Webpage](#) | Essential

3. Debi Andrus's review of *Inside the Tornado*, published in *The Journal of Marketing*.

Reviewed Work: Inside the Tornado: Marketing Strategies from Silicon Valley's Cutting Edge by Geoffrey A. Moore - in *Journal of Marketing*, by Debi Andrus, April 1997

[Article](#) | Essential

4. The Preface and Chapters 1, 3, and 7 of Elliot Rogers' *Diffusion of Innovations*. E-book in UoB Library:

Diffusion of innovations, by Everett M. Rogers, 2003

[Book](#) | Essential

5. Everett Rogers giving a lecture on his work (you can skip forwards over the first 7 minutes which is a rambling preamble: the actual content lasts 43m40s).

Everett Rogers "Diffusion of Innovations" Speech, by Everett Rogers, 30 May 2014

[Webpage](#) | Essential

6. Finally, please read all of V. Govindarajan & C. Trimble (2010) *The Other Side of Innovation: Solving the Execution Challenge*, published by Harvard Business Review Press (this book should take about 5 hours to read in total).

The other side of innovation : solving the execution challenge, by Vijay Govindarajan; Chris Trimble, 2010

[Book](#) | Essential

EMATM0049 (TIBS) Theme 3: Organizing and Financing Innovation (Teaching block 1 2025/26)

[View Online](#)

In this theme we explore two questions:

- (1) Are centralised research labs the best way for a corporation to foster innovation?
- (2) If you want to innovate, but you need financing to do it, where does that money come from -- government funding or start-up investing?

For the first question, we explore thinking around the best way to organize innovation. In particular, we'll look at one of the big shifts in the past twenty years, a tension between centralised innovation (taking place in major research labs) and the idea of open innovation, which was popularized by Henry Chesbrough. To contextualise Chesbrough's work, it is useful to have some knowledge of long-established centralised corporate technology research labs such as Xerox PARC (Palo Alto Research Centre), Bell Labs, and Hewlett-Packard Labs – all of which are now either closed or greatly scaled back from their heyday. Chesbrough's highly influential work on Open Innovation, where research-and-development is largely replaced by acquisition-and-development, arguably hastened the demise of the grand corporate research labs – major corporations switched from funding their own research labs to instead selectively acquiring promising start-ups. While we're looking at these issues, we'll draw upon books by Chesbrough (2003, 2019); Gertner (2013); Hiltzik (2000); and House & Price (2009).

For the second question, we turn to the issue of financing innovation. Here we'll look at where the money comes from, and where it goes; we'll look at Mariana Mazucatto's highly influential writings on the role of state-funding in the innovation process; we'll look at how to form a start-up and what to do once it's in business; and we'll look at two contrasting views on how best to build a company focused on innovation. For this we will explore arguments made in the following books: Mazucatto (2013, 2019); Hoffman & Yeh (2018); Kawasaki (2009, 2015); and Reis (2011).

46 items

Open innovation : the new imperative for creating and profiting from technology, by Henry

William Chesbrough, 2003

[Book](#) | Recommended

Open innovation results : going beyond the hype and getting down to business, by Henry William Chesbrough, 2019

[Book](#) | Recommended

The idea factory : Bell Labs and the great age of American innovation, by Jon Gertner, 2012

[Book](#) | Recommended

Dealers of lightning : Xerox PARC and the dawn of the computer age, by Michael A. Hiltzik, 1999

[Book](#) | Recommended

Blitzscaling : the lightning-fast path to building massively valuable businesses, by Reid Hoffman; Chris Yeh, 2018

[Book](#) | Recommended

The HP Phenomenon : Innovation and Business Transformation, by Charles H. House; Raymond L. Price, 2009

[Book](#) | Recommended

The art of the start : the time-tested, battle-hardened guide for anyone starting anything, by Guy Kawasaki; Paul Boehmer; Tantor Media, 2009

[Document](#) | Recommended

The art of the start 2.0 : the time-tested, battle-hardened guide for anyone starting anything, by Guy Kawasaki; Paul Boehmer; Tantor Media, 2015

[Document](#) | Recommended

The entrepreneurial state : debunking public vs. private sector myths, by Mariana Mazzucato, 2013

[Book](#) | Recommended

The value of everything : making and taking in the global economy, by Mariana Mazzucato, 2018

[Book](#) | Recommended

The lean startup : how constant innovation creates radically successful businesses, by Eric Ries, 2011

[Book](#) | Recommended

As usual, we don't expect you to read all of those books in one week of study, but we do expect you to work in sequence through the following priority-ordered lists of learning materials, spending up to 12 hours on reading or watching these resources, as preparation for the round-table discussions in the seminar sessions. Note that for this theme we have two priority-ordered lists: please work through the two in parallel -- that is, don't work through all of one and ignore the other!

Centralised Laboratories vs Open Innovation

The first four resources introduce three major and long-lasting corporate research labs: Bell Labs; Xerox PARC; and HP Labs. Then resources 5 and 6 introduce open innovation.

1. The brief Introduction: Wicked Problems section that starts The Ideas Factory gives you some idea of the significance of Bell Labs:

The idea factory : Bell Labs and the great age of American innovation, by Jon Gertner, 2012

[Book](#) | Essential

2. A short video about Xerox PARC and the influence the Xerox Alto had on Steve Jobs and Bill Gates (8m01s):

The Xerox Thieves: Steve Jobs & Bill Gates, 31 March 2018

[Webpage](#) | Essential

3. Chapter 4 of The HP Phenomenon describes the creation of HP Labs in 1966:

The HP Phenomenon : Innovation and Business Transformation, by Charles H. House; Raymond L. Price, 2009

[Book](#) | Essential

4. A very brief video of Steve Wozniak, cofounder of Apple, reflecting on his time at Hewlett-Packard (2m17s):

Apple Co-Founder 'Strongly Influenced' by the HP of Yesterday, 29 September 2010

[Webpage](#) | Essential

5. Video of Henry Chesbrough giving a lecture summarising his own work in 2016 (48m35s):

Open Innovation — A New Model for Industrial Innovation, by Henry Chesbrough, 12 October 2016

[Webpage](#) | Essential

6. E. Enkel, M. Bogers, & H. Chesbrough (2020) Exploring open innovation in the digital age: A maturity model and future research directions, R&D Management 50(1):161-168:

Exploring open innovation in the digital age: A maturity model and future research directions, by Ellen Enkel; Marcel Bogers; Henry Chesbrough, January 2020

[Webpage](#) | Essential

Financing Innovation

The first three resources introduce Marianna Mazzucato's arguments in The Entrepreneurial State, along with a critique of those arguments from Stian Westlake. One way of financing innovation is to form a start-up company and attract investors, so the next set of resources cover Eric Ries's hugely influential 2011 book The Lean Startup, and introduce Guy Kawasaki whose 2004 book The Art of the Start is cited as a major influence by many founders of successful technology companies. Finally we explore the thinking of Reid

Hoffman (founder of LinkedIn) on his favoured approach to growing a successful startup -- an approach that Hoffman refers to as Blitzscaling.

1. This brief July 2017 article by Rutger Bregman writing in The Guardian nicely explains Mazzucato's core thesis in her book The Entrepreneurial State, and some of the readers' comments below the article nicely illustrate the extent to which the argument is entwined with political viewpoints:

Look at the phone in your hand - you can thank the state for that | Rutger Bregman, by Rutger Bregman, 2017-07-12

[Webpage](#) | Essential

2. Mariana Mazzucato giving a TED talk on her work: only 14m but it covers the core arguments:

Government -- investor, risk-taker, innovator, by Mariana Mazzucato, 28 October 2013

[Webpage](#) | Essential

3. Stian Westlake writing for Nesta (The UK National Endowment for Science Technology and the Arts) where he is interestingly critical of Mazzucato's arguments in the Entrepreneurial State:

Some thoughts on The Entrepreneurial State, by Stan Westlake; Stian Westlake, 10 November 2014

[Webpage](#) | Essential

4. Nice animated video summary of Kawasaki's The Art of the Start (2m45s):

"The Art of The Start 2.0" by Guy Kawasaki - VIDEO BOOK SUMMARY, by Snackable Idea, 20 March 2015

[Webpage](#) | Essential

5. Guy Kawasaki TED talk: The Art of Innovation. (21m15s)

TED Talk: The Art of Innovation, by Guy Kawasaki, 23 February 2014

[Webpage](#) | Essential

6. An interview conversation between Eric Ries and Jim Euchner summarises the key points of The Lean Startup: E. Ries & J. Euchner (2013) What Large Companies Can Learn From Startups. Research-Technology Management 56(4):12-16:

Conversations: What Large Companies Can Learn from Start-ups: An Interview with Eric Ries - in Research Technology Management, by Eric Ries; Jim Euchner, 2013

[Journal](#) | Essential

7. Eric Ries, author of The Lean Startup, talking about his book (pre-publication) in a one-hour seminar at Google in 2011 (58m09s):

The Lean Startup, by Eric Ries, 7 April 2011

[Webpage](#) | Essential

8. April 2016 article in the Harvard Business Review on Reid Hoffman and Blitzscaling (5 pages):

Blitzscaling - in Harvard business review, by Reid Hoffman; Tim Sullivan, 2016

Article | **Essential**

9. Reid Hoffman in conversation about Blitzscaling with Microsoft CEO Satya Nadella (1h08m54s):

Blitzscaling with Microsoft CEO Satya Nadella and Greylock Partner Reid Hoffman, by Reid Hoffman; Satya Nadella, 14 November, 2018

Webpage | **Essential**

10. A recent and comprehensive academic journal-paper on blitzscaling: D. F. Kurato, H. L. Holt, & E. Neubert (2020) Blitzscaling: The good, the bad, and the ugly. Business Horizons, 63:109-119 (10 pages):

Blitzscaling: The Good, The Bad, and The Ugly - in Business Horizons, by Donald Kuratko; Harrison Holt; Emil Neubert, January 2020

Article | **Essential**

EMATM0049 (TIBS) Theme 4: Legal Matters (Teaching block 1 2025/26)

[View Online](#)

In our fourth theme, we move away from the details of innovation and look instead at legal issues that any data scientist should be aware of. In particular, we look at intellectual property rights (patents, copyright, trade-secrets, and open-source); we cover the European Union's General Data Protection Regulation (GDPR) which has been influential in framing data-protection legislation around the world; and we also explore current thinking on the European Commission's "Artificial Intelligence Act" (technically, Proposal for the Regulation of the European Parliament and of the European Council Laying Down Harmonised Rules on Artificial Intelligence and Amending Certain European Union Legislative Acts).

77 items

Sources that we draw from in this theme include the following books:

Working in public : the making and maintenance of open source software, by Nadia Eghbal, 2020

[Book](#) | Recommended

Intellectual Property, by David Bainbridge, 2018

[Book](#) | Recommended

Information Technology and Intellectual Property Law, by David Bainbridge, 2019

[Book](#) | Recommended

GDPR for Dummies, by Suzanne Dibble, 2020

[Book](#) | Recommended

The cathedral and the bazaar : musings on Linux and open source by an accidental revolutionary, by Eric S. Raymond, 2001

[Book](#) | Recommended

AI engineers's guide to surviving the EU AI Act : navigating the EU regulatory requirements, by Larysa Visengeriyeva, 2025

[Book](#) | Recommended

Learning Materials

Before the seminar discussion session, please work your way through this priority-ordered list of readings and videos, in sequence, and stop when you have spent 12 hours on this theme. The videos add up to less than two hours of total viewing time.

1. Channel 4 News segment explaining GDPR in under 6 minutes (5m39s):

GDPR explained: How the new data protection act could change your life, 23 May 2018

[Webpage](#) | Essential

2. UK-IPO video explanation of UK patents in under 6 minutes (5m34s):

Patents, by Mark Lewney, 31 July 2014

[Webpage](#) | Essential

3. The UK Intellectual Property Office (IPO) has a YouTube channel with lots of useful explanatory videos, all of which are very short (

3.1 This is a general introduction to what the IPO covers (1m20s):

IP Basics: What is Intellectual Property?, 27 November 2015

[Webpage](#) | Essential

3.2 Feel free to browse liberally around the UK-IPO's YouTube channel, but please be sure to watch the following:

3.2a "Masterclass" Intro/Welcome video (1m50s):

Introduction to the UK IPO Masterclass, 31 July 2014

[Webpage](#) | Essential

3.2b "Masterclass on IP" -- this is a video advert for a three-day online course but it explains **why** you'd want to do that course (1m25s):

IPO Masterclass, 12 August 2014

[Webpage](#) | Essential

3.2c "Masterclass on Patents" (2m17s):

Patents Master Class, 31 July 2014

[Webpage](#) | Essential

3.2d "Masterclass on Trademarks" (2m41s):

Trademarks Master Class, 31 July 2014

[Webpage](#) | Essential

3.2e "Masterclass on Designs" (2m21):

Designs Master Class, 31 July 2014

[Webpage](#) | Essential

4. Video by the US Patent and Trademark Office (US-PTO) "What is the Public Domain?" (3m24s)

What is Public Domain?, 18 December 2019

Webpage | **Essential**

5. A nice animation video on Creative Commons (5m32s):

Creative Commons Licenses Explained, 12 October 2011

Webpage | **Essential**

6. J. Chang, et al. (2023) The EU's AI Act, Explained. Wall Street Journal Tech News Briefing video (5m02s):

The EU's AI Act Explained, by Julie Chang, 20 June 2023

Webpage | **Essential**

7. K. Ferguson, et al. (2023) European Parliament Debates Proposed Law on AI Regulation. Deutsche Welle (DW) News video (6m38s):

European Parliament Debates Proposed Law on AI Regulation, by Katie Ferguson, 13 June 2023

Webpage | **Essential**

8. Two descriptions of the same idea, an idea that gave birth to the trillion-dollar company called Alphabet (previously Google):

8a. S. Brin & L. Page (1998) The anatomy of a large-scale hypertextual Web search engine. Computer Networks and ISDN Systems. 30(1-7):107-117:

The Anatomy of a Large-Scale Hypertextual Web Search Engine - in Computer Networks and ISDN Systems, by Sergey Brin; Lawrence Page

Article | **Essential**

8b. L. Page (1998) Method for Node Ranking in a Linked Database. US Patent Number 6285999:

US Patent: Method for Node Ranking in a Linked Database, by Lawrence Page, 9 January 1998

Legal Document | **Essential**

9. D. Zhang, et al. (2023) Decoding the EU Artificial Intelligence Act. Stanford Human-Centered AI Video: a panel discussion among experts (1h04m03s):

Decoding the EU Artificial Intelligence Act, by Daniel Zhang, 6 July 2023

Webpage | **Essential**

10. N. Smuha, et al. (2021) How the EU can Achieve Legally Trustworthy AI: A Response to the European Commission's Proposal for an Artificial Intelligence Act. Technical Report from the Legal, Ethical, & Accountable Digital Society (LEADS) Lab at the University of Birmingham Law School, UK.

How the EU Can Achieve Legally Trustworthy AI: A Response to the European Commission's Proposal for an Artificial Intelligence Act, by Nathalie Smuha; Emma Rengers; Adam Harkens; Wenlong Li; James MacLaren; Riccardo Piselli; Karen Yeung, 5

August 2021

[Article](#) | **Essential**

11. Materials on Faccenda Chicken vs Fowler:

11a. Swarb:

Faccenda Chicken Ltd v Fowler: CA 1986, 10 November 2021

[Webpage](#) | **Essential**

11b. Blackstone:

Defining the Indefinable, by Robert Howe, 24 March 2011

[Document](#) | **Essential**

12. Brief articles on Vestergaard Frandsen A/S and others v Bestnet Europe Ltd:

12a. Coopers:

Legal Updates, Trade Secrets: Vestergaard Frandsen vs Bestnet Europe Ltd, 2013

[Website](#) | **Essential**

12b. D. Cran & L. Morgan (2013):

Vestergaard – guarding trade secrets, by David Cran; Louise Morgan, 29 July 2013

[Webpage](#) | **Essential**

13. M. Borghi (2012) A black and white photo of a red bus. Kluwer Copyright Blog:

A black and white photograph of a red bus, by Maurizio Borghi, 13 March 2012

[Website](#) | **Essential**

Background Material

None of the following materials are required for the seminar discussion, but are listed here for those students who would like to explore legal matters in more depth...

L. Floridi, et al. (2023) capAI: A Procedure for Conducting Conformity Assessment of AI Systems in line with the EU Artificial Intelligence Act. Technical Report, Säid Business School, University of Oxford. Available at ssrn.com/abstract=4064091. A nice Medium article by Elizabeth Obee from August 2021, on the rise of patent applications relevant to data science

(US-centric): <https://towardsdatascience.com/artificial-intelligence-patent-101-3eebf93f5297> Chapter 3 (pp.65-96) of S. Scotchmer's Innovations and Incentives (MIT Press, 2004) is a good short overview of American intellectual property law for the non-lawyer; you can use this for reference purposes. US Patent and Trademark Office: www.uspto.gov UK Intellectual Property Office: www.ipo.gov.uk

This comprehensive guide to GDPR was released by top London IP lawyers Bird & Bird LLP:

General Data Protection Regulation (GDPR) Guide

[Document](#) | **Further**

On the IP of open-source:

The cathedral and the bazaar : musings on Linux and open source by an accidental revolutionary, by Eric S. Raymond, 2001

[Book](#) | [Further](#)

The Promise of Research on Open Source Software, by Georg von Krogh; Eric von Hippel, 2006

[Document](#) | [Further](#)

Working in public : the making and maintenance of open source software, by Nadia Eghbal, 2020

[Book](#) | [Further](#)

Open source software : the new intellectual property paradigm, by Stephen M. Maurer; Suzanne Scotchmer; National Bureau of Economic Research, 2006

[Book](#) | [Further](#)

On the IP of open-sauce:

Norms-Based Intellectual Property Systems: The Case of French Chefs - in Organization Science, by Emmanuelle Fauchart; Eric von Hippel, 2008

[Article](#) | [Further](#)

Kitchen confidential? Norms for the use of transferred knowledge in gourmet cuisine - in Strategic Management Journal, by Giada Di Stefano; Andrew A. King; Gianmario Verona, 2014

[Article](#) | [Further](#)

Why it's so hard to own a recipe, by Adam Ragusea, 2 November 2020

[Webpage](#) | [Further](#)

The final judgements in the lawsuits we explore in this theme:

Court of Appeal: Judgement on Faccenda Chicken vs Fowler, by Lord Kerr; Lord Neill; Lord Nourse, 5 December 1985

[Document](#) | [Further](#)

Vestergaard Frandsen A/S (now called MVF3 APS) and others (Appellants) v Bestnet Europe Limited and others (Respondents), by Lord Neuberger; Lord Clarke; Lord Sumption; Lord Reed; Lord Carnwath, 22 May 2013

[Document](#) | [Further](#)

Judgement between Temple Island Collections Ltd (Claimant) --and-- (1) New English Teas Ltd (2) Nicholas John Houghton (Defendants), by Judge Birss, 28 November 2011

[Document](#) | [Further](#)

On the implications of Generative AI for copyright and patents:

How Generative AI Turns Copyright Law Upside Down - in Science and Technology Law Review, by Mark Lemley, 2024

[Article](#) | [Further](#)

AI-Generated Inventions: Implications for the Patent System, by Gaétan de Rassenfosse; Adam Jaffe; Melissa Wasserman, 2 December 2023

[Document](#) | [Further](#)

The text of the EU AI Act:

EU AI Act: (1) Main text; (2) Annex 1, 21 April 2021

[Webpage](#) | [Further](#)

EMATM0049 (TIBS) Theme 5: Monopolies, Oligopolies, and Anti-Trust. (Teaching block 1 2025/26)

View Online



Theme 5 continues to explore legal issues and regulation, but here we focus on national/international policy-level questions around the laws concerning monopolies and oligopolies, known in the USA as anti-trust laws, and the extent to which lax enforcement of those laws has led to fundamental shifts in the structure of advanced western economies, and the ability of their governments to regulate them. We start with a very short introduction to these laws by Ariel Ezrachi (2021); then we explore the arguments of influential thinkers/writers Cory Doctorow (Giblin & Doctorow, 2022; Doctorow 2024, 2025) and Yanis Varoufakis (2024).

41 items

The internet con : how to seize the means of computation, by Cory Doctorow, 2023

[Book](#) | Recommended

ENSHITTIFICATION : why everything suddenly got worse and what to do about it, by CORY DOCTOROW, 2025

[Book](#) | Recommended

Competition and antitrust law : a very short introduction, by Ariel Ezrachi, 2021

[Book](#) | Essential

Chokepoint Capitalism: how big tech and big content captured creative labour markets, and how we'll win them back, by Rebecca Giblin; Cory Doctorow, 10 November 2022

[Book](#) | Recommended

Technofeudalism : what killed capitalism, by Yanis Varoufakis, 2023

[Book](#) | Recommended

To understand what is argued in these books, please work your way through the following priority-ordered list of learning resources. Note that the first resource is an entire book which you are required to read, but it should take you less than three hours to do so.

1. Ariel Ezrachi's 2021 book "Competition and Anti-Trust Law: a very short introduction" tells you everything you need to know for the purposes of this theme: it should take you less than three hours to read:

Competition and antitrust law : a very short introduction, by Ariel Ezrachi, 2021

[Book](#) | Essential

2a. This short summary article by Cory Doctorow in The Guardian summarises the core arguments of his most recent book, 2025's Enshittification:

Way past its prime: how did Amazon get so rubbish? - in The Guardian, by Cory Doctorow, 2025-10-05, 5 October 2025

[Article](#) | **Essential**

2b. Kyle Chayka writing in The Atlantic gives his brief summary of Enshittification, ending with some optimism:

The Age of Enshittification - in The New Yorker, by Kyle Chayka, 2025

[Article](#) | **Recommended**

3. This longer-form (56m59s) video discussion between Tim O'Reilly (founder of O'Reilly technical book publishers) and Cory Doctorow goes into more detail about enshittification, but it opens with 5 minutes of Doctorow talking about his new fiction book "The Bezzle" which is a collection of stories about the world of Tech and Silicon Valley: the real conversation starts at approx 5m05s:

Tim O'Reilly and Cory Doctorow on "enshittification" and the future of AI : how do we avoid the race to the bottom with AI?, by Tim O'Reilly; Cory Doctorow; O'Reilly (Firm), 2024

[Audio-visual document](#) | **Essential**

4. This brief (less than 2 pages) editorial by Toomas Timpka in The Journal of Science and Medicine in Sport discusses the problem of enshittification in managing academic journals -- what he writes here is applicable to any and all scientific/academic journals, not just journals on sport medicine:

The “enshittification” of online information services obligates rigorous management of scientific journals - in Journal of Science and Medicine in Sport, by Toomas Timpka

[Article](#) | **Essential**

5. This 16-page paper from June 2025 by Michael Maffie and Hector Hurtado in The British Journal of Industrial Relations explores in depth the enshittification of work in the gig economy:

The Enshittification of Work: Platform Decay and Labour Conditions in the Gig Economy - in British Journal of Industrial Relations, by Michael David Maffie; Hector Hurtado, 2025/08/05

[Article](#) | **Essential**

6. Yanis Varoufakis served as the Minister for Finance in the Greek government during the global financial crisis. His 2024 book Technofeudalism is a left-wing blast at Big Tech, very eloquently argued. Here is a 16-minute video that Varoufakis made to summarise his book:

Technofeudalism: the video (short), by Yanis Varoufakis, 4 February 2024

[Webpage](#) | **Essential**

7. Here (hosted on Varoufakis's own website) are two reviews of Technofeudalism from politically right-of-centre reviewers:

Two reviews of TECHNOFEUDALISM by Conservative Publications, by Carlos Perona Calvete; Robert Ballfiore; Yanis Varoufakis, 1 April 2024

[Webpage](#) | **Essential**

8. A longer (49m57s) talking-head video of Varoufakis explaining his arguments in Technofeudalism:

Yanis Varoufakis EXPOSED How Tech Lords Enslaved Humanity, by Yanis Varoufakis, 13 May 2025

[Webpage](#) | Essential

9. Nicholas Gane's 2024 review of Technofeudalism for the Journal of Classical Sociology:

Capitalism is capitalism, not technofeudalism - in Journal of Classical Sociology, by Nicholas Gane, 2024

[Article](#) | Essential

10. Oli Sharpe's review of Technofeudalism, a video for Sharpe's Go Meta Youtube channel:

Dr Oli Sharpe's "Go Meta" video review of "Technofeudalism", by Oli Sharpe, 12 May 2024

[Webpage](#) | Essential

11. Anthony Pereira's review of Technofeudalism written for Human Rights Review:

Technofeudalism: What Killed Capitalism by Yanis Varoufakis Brooklyn, NY: Melville House, 2023 - in Human Rights Review, by Anthony W. Pereira, 2024

[Article](#) | Essential

12. Prior to Enshittification, Cory Doctorow's previous non-fiction book was "The Internet Con: How to Seize the Means of Computation", which is reviewed here by Tim Ribaris writing for the Surveillance and Society journal in 2024:

Review of Doctorow's The Internet Con: How to Seize the Means of Computation - in Surveillance & Society, by Tim Ribaric, 2024

[Article](#) | Essential

13. Here is a conversation about The Internet Con that Doctorow had with Jesse Thorn, host of the Bullseye podcast, in early 2024. This is audio-only but a written transcript is available from the website:

Bullseye (podcast) with Jesse Thorn: Cory Doctorow on The Internet Con, by Jesse Thorn; Cory Doctorow, 29 January 2024

[Webpage](#) | Essential

The seeds of Cory Doctorow's current arguments about internet platform decay (or "enshittification") can be found in the 2022 book that he co-authored with Rebecca Giblin, Chokepoint Capitalism. Here are three brief reviews/summaries of that book:

14. Chokepoint Capitalism reviewed in 2024 by Guy Morrow for the International Journal of Music Business Research:

Chokepoint Capitalism: How Big Tech and Big Content Captured Creative Labour Markets and How We Will Win Them Back - in International Journal of Music Business Research, by Guy Morrow, 2024

[Article](#) | Essential

15. Oli Sharpe's review of Chokepoint Capitalism, a video for Sharpe's Go Meta Youtube channel:

Dr Oli Sharpe's "Go Meta" video review of "Chokepoint Capitalism", by Oli Sharpe, 13 September 2024

[Webpage](#) | Essential

16. Marco Schirone's review of Chokepoint Capitalism, written for the Journal of Librarianship and Scholarly Communication:

Brief Reviews of Books and Products: Chokepoint capitalism - in Journal of Librarianship and Scholarly Communication, by Marco Schirone, 2023

[Article](#) | Essential

EMATM0049 (TIBS) Theme 6: Responsibility and Ethics

(Teaching block 1 2025/26)

[View Online](#)


While it's reasonable to presume that most data scientists would want to operate within the law, assuming those laws are themselves reasonable, merely adhering to the law often leaves quite a lot of latitude, quite a big range of possible behaviours or ways of acting, all of which are legal, but some of which are less responsible, or less ethical, than others. For this reason, our sixth theme is Responsibility & Ethics where among other things we look at the notion of responsible research & innovation (RRI), which has been taken to heart by a number of major organizations in recent years. The books listed for this theme are better treated as background reading: instead in the seminar-preparation readings we will concentrate on peer-reviewed academic papers, which can be pretty heavy going -- so there's a reduced reading list this for this theme, and a few explanatory videos to help lighten the load.

43 items

Books that are relevant to this theme include the following:

Ethics in a computing culture, by William John Brinkman; Alton F. Sanders, 2013

[Book](#) | Recommended

Engineering ethics : concepts and cases, by C. E. Harris; Michael S. Pritchard; Ray W. James; Elaine E. Englehardt; Michael J. Rabins, 2009

[Book](#) | Recommended

The ethical algorithm : the science of socially aware algorithm design, by Michael Kearns; Aaron Roth, 2020

[Book](#) | Recommended

Data science ethics : concepts, techniques, and cautionary tales, by David Martens, 2022

[Book](#) | Recommended

Strategies for responsible innovation : perspectives, projects, and tools for co-creating the future we want, by Andre

Martinuzzi; Florian Findler; Alexander Graf, 2025

[Book](#) | Recommended

Responsible Innovation : Managing the Responsible Emergence of Science and Innovation in Society, by Richard Owen; J. R. Bessant; Maggy Heintz, 2013

[Book](#) | Recommended

Here is the priority-ordered list of learning resources for this theme. As always, please

start at item (1) and work your way down the list in sequence, stopping when you reach the end of the list, or have spent 12 hours studying for this theme -- whichever comes sooner:

(1) UoB's Prof Richard Owen, a vocal proponent of RRI, talking in 2015 (when he was working at Exeter Uni), here giving a brief explanation of what RRI is about:

Responsible Research and Innovation: Why? What is it? with Richard Owen, by Richard Owen, 13 August 2015

[Webpage](#) | Essential

(2) A brief animation explaining how to apply RRI, focused on e-health companies but the broader message here is not domain-specific:

How to apply Responsible Research and Innovation (RRI) in industry – an interactive guide, 10 June 2016

[Webpage](#) | Essential

(3) Michael Kearns & Aaron Roth doing a two-hander lecture at Google in late 2019, talking about The Ethical Algorithm:

The Ethical Algorithm | Michael Kearns & Aaron Roth | Talks at Google, by Michael Kearns; Aaron Roth, 18 December 2019

[Webpage](#) | Essential

(4) Michael Kearns explaining differential privacy to Lex Fridman:

Michael Kearns: Differential Privacy, by Lex Fridman; Michael Kearns, 21 November 2019

[Webpage](#) | Essential

(5) Latanya Sweeney's seminal paper on k-anonymity:

k-Anonymity: A model for protecting privacy - in International Journal of Uncertainty, Fuzziness and Knowledge-Based Systems, by Latanya Sweeney, 2002

[Document](#) | Essential

(6) The famous "Stochastic Parrots" paper by Bender, Gebru, McMillan-Major, and Shmitchell, published in 2021:

On the Dangers of Stochastic Parrots: Can Language Models Be Too Big?, by Emily Bender; Timnit Gebru; Angelina McMillan-Major; Shmargaret Shmitchell, 2021

[Document](#) | Essential

(7) Vern McKinley's 2020 review of The Ethical Algorithm in the Regulation journal:

The Ethical Algorithm - in Regulation, by Vern McKinley, 2020

[Article](#) | Essential

(8) A. Domínguez Hernández, R. Owen, D. Saattrup Nielsen, & R. McConville (2023) Ethical, political and epistemic implications of machine learning (mis)information classification: insights from an interdisciplinary collaboration between social and data scientists. Journal of Responsible Innovation, 10:1, 2222514,

Ethical, political and epistemic implications of machine learning (mis)information classification: insights from an interdisciplinary collaboration between social and data scientists - in Journal of Responsible Innovation, by Andrés Domínguez Hernández; Richard Owen; Dan Saattrup Nielsen; Ryan McConville, 2023

[Document](#) | [Essential](#)

Background Reading

A selection of short contemporary media articles about the way Google treated Timnit Gebru, and on Frances Haugen's whistleblowing of Facebook...

B.1: Timnit Gebru and Google

(B.1.1) D. Harwell & N. Tiku (2020) Google's star AI ethics researcher, one of few black women in the field, says she was fired for a critical email. The Washington Post, 3rd December 2020.

Google's star AI ethics researcher, one of a few Black women in the field, says she was fired for a critical email - in The Washington Post, by Drew Harwell; Nitasha Tiku, 3 December 2020

[Article](#) | [Further](#)

(B.1.2) Casey Newton writing for Platformer reproduces the emails between Gebru and Google:

The withering email that got an ethical AI researcher fired at Google - in Platformer, by Casey Newton, 3 December 2020

[Article](#) | [Further](#)

(B.1.3) MIT Technology Review article by Karen Hao:

We read the paper that forced Timnit Gebru out of Google. Here's what it says - in MIT Technology Review, by Karen Hao, 4 December 2020

[Article](#) | [Further](#)

(B.1.4) A retired academic enters the debate...

Retired UW computer science professor embroiled in Twitter spat over AI ethics and 'cancel culture' - in GeekWire.com, by Taylor Soper, 16 December 2020

[Webpage](#) | [Further](#)

(B.2) Francis Haugen and Facebook

(B.2.1) J. Waterson and D. Milmo (2021) Facebook whistleblower Frances Haugen calls for urgent external regulation. The Guardian, 25th October 2021:

Facebook whistleblower Frances Haugen calls for urgent external regulation - in The Guardian, by Jim Waterson; Dan Milmo, 2021-10-25

[Article](#) | [Further](#)

(B.2.2) D. Milmo (2021) Frances Haugen: 'I never wanted to be a whistleblower. But lives

were in danger'. The Guardian, 24th October 2021:

Frances Haugen: 'I never wanted to be a whistleblower. But lives were in danger' - in The Guardian, by Dan Milmo, 24 October 2021

[Article](#) | [Further](#)

(B.2.3) D. Milmo (2021) Mark Zuckerberg should quit Facebook says Frances Haugen: The Guardian, 1st November 2021:

Mark Zuckerberg should quit Facebook, says Frances Haugen - in The Guardian, by Dan Milmo, 1 November 2021

[Article](#) | [Further](#)

(B.2.4) J. Walsh (2021) Why Facebook's Metaverse is Dead on Arrival. The Intelligencer, 8th November 2021:

Why Facebook's Metaverse Is Dead on Arrival - in New York Magazine "Intelligencer", by James Walsh, 8 November 2021

[Article](#) | [Further](#)

EMATM0049 (TIBS) Theme 7: Technology Failures (Teaching block 1 2025/26)

[View Online](#)

Even if you act within the law, and ethically and responsibly, it's plausible that the technology you are involved in working on is "risky", in the sense that it's not guaranteed to be failure-proof, and maybe you can't always state in advance all the ways in which it might fail, and what the consequences might be. This notion, of failures of risky technologies, has been studied by sociologists of science and technology, and that is what we look at in our seventh theme: Technology Failures, where the books we draw on include: Perrow (1984); Petroski (1985); Vaughan (1996); Starbuck & Farjoun (2005); Reason (2008); McDonald & Hansen (2009); and Petroski (2018)...

59 items

Truth, lies, and O-rings : inside the space shuttle Challenger disaster, by Allan J. McDonald; James R. Hansen, 2009

[Book](#) | Recommended

Normal accidents : living with high-risk technologies, by Charles Perrow, 1984

[Book](#) | Recommended

To engineer is human : the role of failure in successful design, by Henry Petroski, 1985

[Book](#) | Recommended

Success through failure : the paradox of design, by Henry Petroski, 2018

[Book](#) | Recommended

The human contribution : unsafe acts, accidents and heroic recoveries, by J. T. Reason, 2008

[Book](#) | Recommended

Organization at the limit : lessons from the Columbia disaster, by William H. Starbuck; Moshe Farjoun, 2005

[Book](#) | Recommended

The Challenger launch decision : risky technology, culture, and deviance at NASA, by Diane Vaughan, 1996

[Book](#) | Recommended

Content Warning: in this theme we cover catastrophic engineering failures in which multiple people lost their lives; some of the video content for this week includes footage of those events, and of the reactions of people closely involved -- colleagues and friends and relatives -- witnessing those events as they occurred.

Introduction/Overview video by Dave Cliff (25m48s):

TIBS Theme 7: Overview Video, by Dave Cliff, 2025[Webpage](#) | **Essential**

Learning Resources

Below is the priority-ordered list of learning resources for this theme. As usual, please start at item (1) and work your way through items in the list, in sequence, until you get to the end of the list or until you have spent 12 hours reading and watching resources for this list, whichever comes sooner...

(1) Emmy-award winning documentary on the loss of Challenger (50m21s):

Challenger: A Rush To Launch, by Jason Payne, 28 January 2016

[Webpage](#) | **Essential**

(2) Documentary on the loss of Columbia (44m31s):

Space Shuttle Columbia - Falling Star HD, 16 January 2016

[Webpage](#) | **Essential**

(3) Allan McDonald, former Director of the Shuttle Solid Rocket project for Morton Thiokol (the company that supplied the shuttle's solid rocket boosters to NASA), speaking calmly and eloquently about the Challenger loss from his perspective, having been intimately involved in the launch decision: he said they should not launch; they ignored him (20m28s):

Space Shuttle Challenger Disaster: Ethics Case Study No. 1, 18 November 2015

[Webpage](#) | **Essential**

(4) Mike Mullane, who was a NASA shuttle astronaut, talking about normalization of deviance in a lecture to firefighters -- machine-gun delivery, articulate and angry (16m35s):

Mike Mullane - Normalisation of deviance - IAFF - Part 1, by Mike Mullane, 1 August 2015

[Webpage](#) | **Essential**

(5) A. Minkes (1996) review of Vaughan's The Challenger Launch Decision, in Technovation , 16(9):525-526:

The challenger launch decision. Risky technology, culture, and deviance at NASA - in Technovation, by A.L. Minkes, 1996

[Article](#) | **Essential**

(6) J. F. Short Jr. (1996) review of Vaughan's The Challenger Launch Decision, in Contemporary Sociology, 25(6):727-729:

The Challenger Launch Decision: Risky Technology, Culture, and Deviance at NASA - in Contemporary Sociology, by James F. Short Jr; Diane Vaughan, 1996

[Article](#) | **Essential**

(7) M. Farjoun & W. Starbuck (2007) Organizing At and Beyond the Limits. Organization Studies, 28(4):541-566:

Organizing at and Beyond the Limits - in Organization Studies, by Moshe Farjoun; William H. Starbuck, 2007

[Article](#) | [Essential](#)

(8) P. Banerjee & J. Mahoney (2007) review of Starbuck & Farjoun's Organization at the Limit: Lessons from the Columbia Disaster, in Academy of Management Perspectives, 21(4):87-90:

Organization at the Limit: Lessons From the Columbia Disaster **Organization at the Limit: Lessons From the Columbia Disaster** Edited by Starbuck William and Farjoun Moshe, Malden, MA: Blackwell Publishing, 2005 - in Academy of Management Perspectives, by Preeta M. Banerjee; Joseph T. Mahoney, 2007

[Article](#) | [Essential](#)

(9) M. Neufeld's 2010 book-review of McDonald & Hansen's Truth, Lies, and O-Rings: Inside the Space Shuttle Challenger Disaster, in Isis 101(2):452-453, 2010:

Allan J. McDonald; James R. Hansen. Truth, Lies, and O-Rings: Inside the Space Shuttle Challenger Disaster . xix + 626 pp., illus., bibl., index. Gainesville: University Press of Florida, 2009. \$39.95 (cloth) - in Isis, by Michael J. Neufeld, 2010

[Article](#) | [Essential](#)

(10) Charles Perrow lecture at Microsoft Research, circa 2006 (1h23m32s):

The Next Catastrophe: Reducing Our Vulnerabilities to Natural, Industrial, and Terrorist Disasters, by Charles Perrow, 7 September 2016

[Webpage](#) | [Essential](#)

(11) D. L. Simms (1986) review of Normal Accidents, in Technology & Culture, 27(4):903-905:

Normal Accidents: Living with High-Risk Technologies - in Technology and Culture, by D. L. Simms; Charles Perrow, 1986-10

[Article](#) | [Essential](#)

(12) J. Ravetz (1985) review of Normal Accidents, in Futures, 17(3):287-288:

Normal accidents; Living with high-risk technologies - in Futures, by Jerome Ravetz, 1985-6

[Article](#) | [Essential](#)

(13) Brief video where James Reason explains his "Swiss Cheese Model" for explaining technology accidents (4m08s):

Reason explains Swiss Cheese Model, 22 October 2018

[Webpage](#) | [Essential](#)

(14) Even briefer video which is a much nicer explanation of Reason's Swiss Cheese model (0m32s):

Prof James Reason's Swiss Cheese Model, by Gareth Lock, 14 December 2015

[Webpage](#) | [Essential](#)

(15) D. Clark (2011) review of Reason's The Human Contribution, in Journal of Risk Research, 14(1):144-145:

The human contribution: unsafe acts, accidents and heroic recoveries - in Journal of Risk Research, by David M. Clarke, 2011-01

[Article](#) | [Essential](#)

(16) M. Young (2010) review of Reason's The Human Contribution, in Ergonomics, 53(4):586-587:

The Human Contribution: Unsafe acts, accidents and heroic recoveries - in Ergonomics, by Mark S. Young, 2010-04

[Article](#) | [Essential](#)

(17) A. Roland (1986) review of Petroski's To Engineer is Human, in Isis 77(3):562-563.

Review: To Engineer Is Human: The Role of Failure in Successful Design by Henry Petroski - in Isis, by Alex Roland, 1986

[Article](#) | [Essential](#)

(18) E. Kemp (1986) review of Petroski's To Engineer is Human, in Technology & Culture, 27(4):846-848.

To Engineer Is Human: The Role of Failure in Successful Design - in Technology and Culture , by Emory L. Kemp; Henry Petroski, 1986-10

[Article](#) | [Essential](#)

Further Reading

(19) D. Cliff & L. Northrop (2011) The Global Financial Markets: An Ultra-Large-Scale-Systems Perspective. Driver Review DR4, commissioned by the UK Government Office for Science Foresight project The Future of Computer Trading in Financial Markets in 2010; published in revised form in Calinescu & Garlan (2012).

The Global Financial Markets: an Ultra-Large-Scale Systems perspective , by Dave Cliff; Linda Northrop, 2010

[Document](#) | [Further](#)

Large-scale complex IT systems : development, operation and management, 17th Monterey workshop 2012, Oxford, UK, March 19-21, 2012 : revised selected papers, by Monterey Workshop; Radu Calinescu; David Garlan, 2012

[Book](#) | [Further](#)

(20) Diane Vaughan seems to have kept a very low profile. The only YouTube video of her speaking is this keynote lecture at the 2016 Annual Sociology Student Conference. Vaughan opens her lecture by saying this is about her forthcoming book Dead Reckoning: Air Traffic Control, System Effects, and Risk, which was eventually published five years after the lecture, in August 2021. In the lecture she does link Dead Reckoning to her work on the Challenger Launch Decision. This lecture is not visually compelling (just a camera pointing at Vaughan as she speaks at the lectern, reading a paper, with no slides or graphics) and the audio quality is really pretty poor too, but her words are full of insights. [youtube.com/watch?v=uu0tPsb_Wnc&t=1845s](https://www.youtube.com/watch?v=uu0tPsb_Wnc&t=1845s) (1h27m11s)

Diane Vaughan, 2016 Annual Sociology Student Conference | Keynote Speaker, by Diane Vaughan, 2 May 2016

[Webpage](#) | **Recommended**

(21) D. Vaughan (2006) NASA Revisited: Theory, Analogy, and Public Sociology, The American Journal of Sociology, 111(2):353-393:

NASA Revisited: Theory, Analogy, and Public Sociology¹ - in American Journal of Sociology, by Diane Vaughan, 2006

[Article](#) | **Further**

(22) The final slides on the intro/overview video lecture for this theme talk about Edward Tufte's contribution to the narrative on the Columbia accident. Tufte's website is at <https://www.edwardtufte.com/tufte/> and the pamphlet mentioned in the video lecture is: E. Tufte (2006) The Cognitive Style of Powerpoint:

EdwardTufte.com, by Edward Tufte, 2025

[Webpage](#) | **Further**

The cognitive style of PowerPoint : pitching out corrupts within, by Edward R. Tufte, 2006

[Book](#) | **Further**

EMATM0049 (TIBS) Theme 8: Societal Failures (Teaching block 1 2025/26)

[View Online](#)

74 items

Continuing the theme of things that can go wrong and that we should look out for, what if you act legally and ethically and responsibly and your technology doesn't fail, but things still don't go well because of a failure not in the technology but in the social context that the technology is deployed in? We should consider not only technology failures that have societal effects, but also societal failures that are enabled by, or caused by, advances in technology. That is, where the tech does not actually fail, but it is misapplied, and its mis-application fails society in some sense. Our eighth theme is technology-enabled societal failures, and we will draw upon the following books: Schneps & Colmez (2013); O'Neill (2016); Bartlett (2018); Moore (2018); Sumpter (2018); Criado Perez (2019); Kaiser (2019); and Wylie (2019).

The people vs tech : how the internet is killing democracy (and how we save it), by Jamie Bartlett, 2018

[Book](#) | Recommended

Invisible women : exposing data bias in a world designed for men, by Caroline Criado-Perez, 2019

[Book](#) | Recommended

Targeted : the Cambridge Analytica whistleblower's inside story of how big data, Trump, and Facebook broke democracy and how it can happen again, by Brittany Kaiser, 2019

[Book](#) | Recommended

Democracy hacked : political turmoil and information warfare in the digital age, by Martin Moore, 2018

[Book](#) | Recommended

Weapons of math destruction : how big data increases inequality and threatens democracy, by Cathy O'Neil, 2017

[Book](#) | Recommended

Math on trial : how numbers get used and abused in the courtroom, by Leila Schneps; Coralie Colmez, 2013

[Book](#) | Recommended

Outnumbered : from Facebook and Google to fake news and filter-bubbles -- the algorithms that control our lives, by David Sumpter, 2018

[Book](#) | Recommended

Mindf*ck : inside Cambridge Analytica's plot to break the world, by Christopher Wylie, 2019

[Book](#) | Recommended

Learning Resources

As usual, please work through this priority-ordered list of learning resources, starting at item (1) and working through them in sequence until either you reach the end of the list or you have spent 12 hours studying these materials -- whichever is sooner. Items (1) to (16) add up to 8h38m35s of total viewing time.

(1) As background on the stories and themes explored in the books by Bartlett, Kaiser, Moore, and Wylie, this TED talk by Observer journalist Carol Cadwalladr compresses much of what you need to know into 16minutes (15m15s). (PS If you have access to Netflix, and if you have the time, then also try to watch The Great Hack, the movie-length documentary on the Cambridge Analytica scandal, in which Carol Cadwalladr also features quite prominently).

Facebook's role in Brexit — and the threat to democracy | Carole Cadwalladr TED 26.8M subscribers Join Subscribe 32K Share, by Carole Cadwalladr, 10 June 2019

[Webpage](#) | Essential

(2) Alexander Nix, CEO of Cambridge Analytica (CA), giving an 11-minute presentation at the Concordia Annual Summit in 2016 (before the scandal broke) where he pitches CA's ability to use behavioral science, big data analytics, and targeted adverts, to influence election outcomes. Here he focuses on CA's successes in helping Senator Ted Cruz rise from relative obscurity to be the last and only serious competition to Donald Trump's nomination in 2016 as the Republican Paty's presidential candidate (11m00s):

The Power of Big Data and Psychographics | 2016 Concordia Annual Summit, by Alexander Nix, 27 September 2016

[Webpage](#) | Essential

(3) BBC Newsnight segment from July 2016 on Cambridge Analytica's role in the EU/Brexit Referendum (13m39s):

Did Cambridge Analytica play a role in the EU referendum?, by Gabriel Gatehouse, 13 July 2017

[Webpage](#) | Essential

(4) Channel4 News segment from October 2016 on use of big data in the 2016 US Presidential Election (9m34s).

Cambridge Analytica: how big data shaped the US election, by Channel 4 News, 2016-10-24T12:41:23-07:00

[Audio-visual document](#) | Essential

(5) Channel4 News segment from March 2018 where they run their Cambridge Analytica expose; includes footage from the secretly-filmed meetings where Channel4 journalists posed undercover as potential new clients of CA. It was all over for CA after this was broadcast (19m12s):

Cambridge Analytica Uncovered: Secret filming reveals election tricks, by Channel 4 News, 2018-03-19T11:47:35-07:00

[Audio-visual document](#) | Essential

(6) Channel4 News segment from March 2018 focused on Christopher Wylie, where he explains the CA slurp of 50million Facebook profiles (16m03s):

Cambridge Analytica: Whistleblower reveals data grab of 50 million Facebook profiles, by Channel 4 News, 2018-03-17T08:05:04-07:00

[Audio-visual document](#) | **Essential**

(7) The Guardian video interview with Christopher Wylie from March 2018 (13m03s):

Cambridge Analytica whistleblower: 'We spent \$1m harvesting millions of Facebook profiles', by The Guardian, 2018-03-17T07:10:02-07:00

[Audio-visual document](#) | **Essential**

(8) Channel4 News segment from June 2018 on CA ex-CEO Alexander Nix being questioned by UK Members of Parliament (11m27s):

Reaction as former Cambridge Analytica CEO is grilled by MPs, by Channel 4 News, 2018-06-06T11:58:32-07:00

[Audio-visual document](#) | **Essential**

(9) Cathy O'Neill lecture at Google, talking about her book Weapons of Math Destruction (58m21s):

Weapons of Math Destruction | Cathy O'Neil | Talks at Google, by Talks at Google, 2016-11-02T11:31:43-07:00

[Audio-visual document](#) | **Essential**

(10) Caroline Criado-Perez interviewed by Helen Lewis at The Fawcett Society (1h15m01s):

Invisible Women: In Conversation with Caroline Criado Perez and Helen Lewis, by The Fawcett Society, 2019-04-29T02:42:53-07:00

[Audio-visual document](#) | **Essential**

(11) Video of David Sumpter lecture at The Royal Institution about The Algorithms that Control Our Lives. For reasons explained in the comments to this video, it appears to start at t=00:00:00 in some random point after the lecture has already begun, but this is the RI's attempt at a teaser/trailer -- the actual lecture begins, with a proper start, 50 seconds into the video (48m52s):

Algorithms That Control Our Lives - with David Sumpter, by The Royal Institution, 2018-07-18T07:03:48-07:00

[Audio-visual document](#) | **Essential**

(12) Coralie Colmez public lecture talking about her book Math on Trial (34m19s):

Coralie Colmez: Maths on Trial, by Question Explore Discover, 2015-01-02T03:28:12-08:00

[Audio-visual document](#) | **Essential**

(13) Interview with Coralie Colmez, one of the two co-authors of Math on Trial (17m08s):

Maths on Trial, by Bond Solon, 2013-10-11T07:19:52-07:00

[Audio-visual document](#) | **Essential**

(14) Jamie Bartlett lecture on The People vs Tech: Bartlett starts speaking 5 minutes into this (57m21s):

#1. People vs Tech - Jamie Bartlett - The Night of Dictatorship 2018, by De Balie, 2018-09-18T14:01:08-07:00

[Audio-visual document](#) | **Essential**

(15) Martin Moore lecture on Democracy Hacked (56:04):

Democracy hacked: how technology is destabilising global politics, by International Journalism Festival, 2019-04-04T02:32:10-07:00

[Audio-visual document](#) | **Essential**

(16) Brittany Kaiser in March 2020 giving a lecture followed by Q&A at The Oxford Union (1h02m16s):

Cambridge Analytica Whistleblower, Brittany Kaiser | Full Address and Q&A | Oxford Union, by OxfordUnion, 2020-03-11T13:00:00-07:00

[Audio-visual document](#) | **Essential**

(18) Margaret McCartney's review of Criado Perez's Invisible Women in The Lancet:

Making women equally visible - in The Lancet, by Margaret McCartney, 2019-06

[Article](#) | **Essential**

(19) Elaine Glaser's review of Criado Perez's Invisible Women in The Guardian:

Invisible Women by Caroline Criado Perez – a world designed for men - in The Guardian, by Eliane Glaser, 2019-02-28

[Article](#) | **Essential**

(20) Angela Saini's review of Criado Perez's Invisible Women in The Observer:

Invisible Women by Caroline Criado Perez – review - in The Guardian, by Angela Saini, 2019-03-11

[Article](#) | **Essential**

(21) A review of Criado Perez's Invisible Women in Forbes, by their senior contributor called GrrlScientist:

Invisible Women: Exposing Data Bias In A World Designed For Men, by Grrrl Scientist

[Document](#) | **Essential**

(22) Emily Bell's review of Bartlett's The People vs Tech in The Guardian:

The People vs Tech by Jamie Bartlett review – once more into the digital apocalypse - in The Guardian, by Emily Bell, 2018-04-16

[Article](#) | **Essential**

(23) Tim Posada's review of Bartlett's The People vs Tech in the journal Cyberspace Studies 3(1):105-107, 2019:

Review of “The People Vs Tech: How the Internet is Killing Democracy (and How We Save It)” by Jamie Bartlett - in Cyberspace Studies, by Tim Posada, 2019/01/01

[Article](#) | [Essential](#)

(24) John Naughton in The Guardian reviewing both Kaiser's Targeted and Wylie's Mindf*ck :

Cambridge Analytica: Mindf*ck by Christopher Wylie; Targeted by Brittany Kaiser – reviews - in The Guardian, by John Naughton, 2019-10-29

[Article](#) | [Essential](#)

(25) Steven Poole's review of Moore's Democracy Hacked in The New Statesman:

Democracy Hacked: how big tech poisoned politics - in The New Statesman, by Steven Poole, 3 October 2018

[Article](#) | [Essential](#)

(26) Mona Chalabi's review of O'Neill's Weapons of Math Destruction in The Guardian:

Weapons of Math Destruction: Cathy O'Neil adds up the damage of algorithms - in The Guardian, by Mona Chalabi, 2016-10-27

[Article](#) | [Essential](#)

(27) Mary Poovey's review of O'Neill's Weapons of Math Destruction in the journal Notices of the American Mathematical Society 64(8):933-935, 2017:

Book Review: Weapons of Math Destruction - in Notices of the American Mathematical Society, by Mary Poovey, September 2017

[Article](#) | [Essential](#)

(28) Daniel Ullman's review of Schneps & Colmez's Math On Trial in the journal The American Mathematical Monthly 121(5):463-466, 2014:

Book Review: "Math on Trial" by Schneps & Colmez - in The American Mathematical Monthly, by Daniel Ullman, May 2014

[Article](#) | [Essential](#)

(29) Pamela Gorkin's review of Schneps & Colmez's Math On Trial in the journal The Mathematical Intelligencer 36(1):78-79, 2014:

Math on Trial: How Numbers Get Used and Abused in the Courtroom by Leila Schneps and Coralie Colmez - in The Mathematical Intelligencer, by Pamela Gorkin, 2014-2

[Article](#) | [Essential](#)

(30) James Bridle's review of Sumpter's Outnumbered in The Guardian:

Outnumbered: From Facebook and Google to Fake News and Filter-bubbles by David Sumpter – review - in The Guardian, by James Bridle, 2018-08-01

[Article](#) | [Essential](#)

(31) Aliya Ram's review of Sumpter's Outnumbered in The Financial Times:

Outnumbered by David Sumpter — can data science really sway polls?, by Aliya Ram, 2018

[Article](#) | Essential

(32) Nate Matias on "Exit-Option Democracy" in Medium:

Quitting Facebook & Google: Why Exit Option Democracy is the Worst Kind of Democracy | by J. Nathan Matias | Medium, by J. Nathan Matias

[Webpage](#) | Essential

Background Reading

E. Douek (2024) The Meta Oversight Board and the Empty Promise of Legitimacy. Forthcoming in the Harvard Journal of Law and Technology, 37:

The Meta Oversight Board and the Empty Promise of Legitimacy - in SSRN Electronic Journal, by Evelyn Douek, 2023

[Article](#) | Further

EMATM0049 (TIBS) Theme 9: Power from Data (Teaching block 1 2025/26)

[View Online](#)

In our ninth theme, Power from Data, we take a look at the extent to which gathering data can be the source of power – either political power, or economic power. We'll see how companies such as Amazon, Facebook, and Google have risen to be global powerhouses, highly dominant and in some cases having essentially monopoly power in their chosen lines of business, by skilfully monetizing data gathered from the users of their services. We'll look at how OpenAI controversially harvested huge amounts of data to train its ChatGPT series of large language models. And finally we'll look at how nation-states, superpowers such as the USA and China, are deploying digital technology to enact large-scale surveillance on their citizens: in each case, the state argues that the surveillance is necessary for the safety of their populace; and in each case, there are voices raised in passionate argument against that surveillance, arguing that it is an imposition on individual liberty. Books that we draw from in this theme include: Greenwald (2014); Taplin (2018); Doctorow (2019); Foroohar (2019); Griffiths (2019); McNamee(2019); Snowden (2019); Chin & Lin (2022); and Hao (2025).

68 items

Surveillance state : inside China's quest to launch a new era of social control, by Josh Chin; Liza Lin, 2022

[Book](#) | Recommended

Don't be evil : the case against big tech, by Rana Foroohar, 2019

[Book](#) | Recommended

No place to hide : Edward Snowden, the NSA, and the surveillance state, by Glenn Greenwald, 2014

[Book](#) | Recommended

The great firewall of China : how to build and control an alternative version of the internet, by James Griffiths, 2019

[Book](#) | Recommended

Empire of AI : inside the reckless race for total domination, by Karen Hao, 2025

[Book](#) | Recommended

Zucked : waking up to the Facebook catastrophe, by Roger McNamee, 2019

[Book](#) | Recommended

Permanent record, by Edward J. Snowden, 2019

[Book](#) | Recommended

Move fast and break things : how Facebook, Google, and Amazon cornered culture and undermined democracy, by Jonathan Taplin, 2017

[Book](#) | Recommended

The age of surveillance capitalism : the fight for a human future at the new frontier of power, by Shoshana Zuboff, 2019

[Book](#) | Recommended

Learning Materials

There are two related, arguably separate, but ultimately linked sub-themes here: Shoshanna Zuboff's notion of surveillance capitalism, where global private-sector corporations such as Facebook or Google are accused of engaging in monetized surveillance of their users; and plain old-fashioned state-surveillance by a government monitoring its citizens, which is made so much easier by digital data-gathering. Here are two priority-ordered lists of material: please sample material from both sub-themes.

(A) Surveillance Capitalism

(A.1) Shoshanna Zuboff documentary by VPRO (Dutch public broadcaster), December 2019 (49m59s):

Shoshana Zuboff on surveillance capitalism | VPRO Documentary, by vpro documentary, 2019-12-20T17:00:04-08:00

[Audio-visual document](#) | Essential

(A.2) Second part of a 2-part interview with Zuboff, but this starts with a recap of Part 1 (22m17s):

Big Tech Stole Our Data While Democracy Slept: Shoshana Zuboff on the Age of Surveillance Capitalism, by Democracy Now!, 2019-03-01T08:48:28-08:00

[Audio-visual document](#) | Essential

(A.3) Zuboff interviewed, and her book reviewed, by John Naughton in The Observer, Oct. 2019:

'The goal is to automate us': welcome to the age of surveillance capitalism - in The Guardian, by John Naughton, 2019-01-20

[Article](#) | Essential

(A.4) Zuboff interviewed by Joanna Kavenna in The Guardian, Oct. 2019:

Shoshana Zuboff: 'Surveillance capitalism is an assault on human autonomy' - in The Guardian, by Joanna Kavenna, 2019-10-04

[Article](#) | Essential

(A.5) An interview between The Observer's Patricia Clarke and Karen Hao, about Hao's 2025 book Empire of AI (57m46s)

Karen Hao interview on her 2025 book "The Empire Of AI", by The Observer UK, 2025-06-12T06:57:15-07:00

[Audio-visual document](#) | Essential

(A.6) Shi En Kim's review of Karen Hao's Empire of AI, for Science News:

Does the AI industry operate like a modern colonial empire?, by Shi En Kim

[Document](#) | **Essential**

(A.7) Cory Doctorow podcast: On The True Dangers of Surveillance Capitalism (58m24s):

Big Tech - S3E01 - Cory Doctorow on the True Dangers of Surveillance Capitalism, by Centre for International Governance Innovation, 2020-11-26T05:00:00-08:00

[Audio-visual document](#) | **Essential**

(A.8) Rana Foroohar interview for Institute of New Economic Thinking (19m32s):

Don't Be Evil. Rana Foroohar on Big Tech., by New Economic Thinking, 2019-11-13T10:17:07-08:00

[Audio-visual document](#) | **Essential**

(A.9) Rana Foroohar interviewed on CBS News (7m01s):

"Don't Be Evil" author on the power of Big Tech, by CBS News, 2019-11-08T11:51:05-08:00

[Audio-visual document](#) | **Essential**

(A.10) Foroohar "Long Read" article in The Guardian, November 2019:

How big tech is dragging us towards the next financial crash - in The Guardian, by Rana Foroohar, 2019-11-08

[Article](#) | **Essential**

(A.11) Interview in The Guardian by Alex Hearn with Roger McNamee, author of Zucked: Waking up to the Facebook Catastrophe:

Roger McNamee: 'It's bigger than Facebook. This is a problem with the entire industry' - in The Guardian, by Alex Hearn, 2019-02-16

[Article](#) | **Essential**

(A.12.1) Part 1 of Roger McNamee interview about his book Zucked (56:59 -- but less when you skip the adverts!):

E908 Roger McNamee, FB investor, author "Zucked": mentoring Zuck, Russia, big data, surveillance-PT1, by This Week in Startups, 2019-03-05T11:25:43-08:00

[Audio-visual document](#) | **Essential**

(A.12.2) Part 2 of Roger McNamee interview about his book Zucked (1h02m10s):

E909 PT2 Roger McNamee FB investor, author "Zucked": SV libertarian cult, Russia, data, surveillance, by This Week in Startups, 2019-03-08T10:42:10-08:00

[Audio-visual document](#) | **Essential**

(A.13) John Harris's review of McNamee's Zucked, in The Guardian:

Zucked by Roger McNamee review - Facebook's catastrophe - in The Guardian, by John Harris, 2019-02-07

[Article](#) | **Essential**

(A.14) David Runciman's review of Taplin's *Move Fast and Break Things*, in *The Guardian*:

Move Fast and Break Things by Jonathan Taplin review – the damage done by Silicon Valley - in *The Guardian*, by David Runciman, 2017-04-26

[Article](#) | **Essential**

(A.15) John Naughton's review of Taplin's *Move Fast and Break Things*, in *The Guardian*:

Move Fast and Break Things review – Google, Facebook and Amazon exposed - in *The Guardian*, by John Naughton, 2017-04-17

[Article](#) | **Essential**

(B) Surveillance by Governments

(B.1) Henry Porter's review in *The Guardian* of Greenwald's *No Place to Hide*:

No Place to Hide review – Glenn Greenwald's compelling account of NSA/GCHQ surveillance - in *The Guardian*, by Henry Porter, 2014-05-19

[Article](#) | **Essential**

(B.2) Glenn Greenwald interview on BBC Newsnight, October 2013 (14m11s):

Glenn Greenwald full interview on Snowden, NSA, GCHQ and spying - Newsnight, by BBC Newsnight, 2013-10-03T15:28:38-07:00

[Audio-visual document](#) | **Essential**

(B.3) A video from the New York Times' Visual Investigations Team, titled "How China's Surveillance is Growing More Invasive" (14m26s).

How China's Surveillance Is Growing More Invasive | Visual Investigations, by The New York Times, 2022-06-22T10:00:23-07:00

[Audio-visual document](#) | **Essential**

(B.4) Edward Snowden interview in *The Guardian*, Sep 2019:

'They wanted me gone': Edward Snowden tells of whistleblowing, his AI fears and six years in Russia - in *The Guardian*, by Ewen MacAskill

[Article](#) | **Essential**

(B.5) Josh Chin and Liza Lin, the two Wall Street Journal (WSJ) journalists who co-wrote the book *Surveillance State*, being interviewed by ex-WSJ journalist Mei Fong, (56m03s):

Josh Chin & Liza Lin, *Surveillance State*: Inside China's Quest to Launch a New Era of Social Control, by New America, 2022-09-19T08:58:20-07:00

[Audio-visual document](#) | **Essential**

(B.6) Ausma Bernot's review for *The British Journal of Criminology* of Chin & Lin's 2022 *Surveillance State*:

Surveillance State: Inside China's Quest to Launch a New Era of Social Control. By Josh

Chin and Liza Lin (St Martin's Press, 2022, 336pp, \$29.99 hbk) - in The British Journal of Criminology, by Ausma Bernot, 2023

[Article](#) | [Essential](#)

(B.7) Lloyd Green's review in The Guardian of Snowden's Permanent Record:

Permanent Record review: Edward Snowden writes in 50 shades of grey - in The Guardian, by Lloyd Green, 2019-09-22

[Article](#) | [Essential](#)

(B.8) Edward Snowden interview on CNBC, Sep. 2019 (54m29s):

Full Interview: Edward Snowden On Trump, Privacy, And Threats To Democracy | The 11th Hour | MSNBC, by MSNBC, 2019-09-17T07:01:10-07:00

[Audio-visual document](#) | [Essential](#)

(B.9) Steven Poole's review in The Guardian of Griffiths's The Great Firewall of China:

The Great Firewall of China by James Griffiths review - how to control the internet - in The Guardian, by Steven Poole, 2019-04-27

[Article](#) | [Essential](#)

(B.10) Jamie Bartlett's two-book review in The Literary Review, where one of the two books is Griffiths's The Great Firewall of China:

The Great Firewall of China: How to Build and Control an Alternative Version of the Internet by James Griffiths; India Connected: How the Smartphone is Transforming the World's Largest Democracy by Ravi Agrawal - review by Jamie Bartlett

[Document](#) | [Essential](#)

(B.11) James Griffiths' lecture on The Great Firewall of China to the Foreign Correspondents Club of Hong Kong (47m11s):

James Griffiths - China's War on the Internet: How the West Lost the Web, by The Foreign Correspondents' Club, Hong Kong, 2019-03-20T01:57:50-07:00

[Audio-visual document](#) | [Essential](#)

(B.12) Michael Anti TED talk Behind the Great Firewall of China (18m51s):

Michael Anti: Behind the Great Firewall of China, by TED, 2012-07-30T08:06:29-07:00

[Audio-visual document](#) | [Essential](#)